DONALD AND BARBARA
ZUCKER SCHOOL *of* MEDICINE
AT HOFSTRA/NORTHWELL®

Structure Lab 1.6: Nervous System Structure

Lab Station & Learning Objectives:

Cranial Nerves and Skull Base

1. Describe the bones that contribute to the anterior, middle, and posterior cranial fossae.
2. Correlate the foramina of the basicranium with the cranial nerves that pass through them.
3. Explain how each cranial nerve contributes to sensation and/or motor function in the head and neck.
4. Predict the effects of lesions to the cranial nerves and how they would manifest on a physical exam.

How to prepare for session:

Prior to the session, identify the following bony features on a skull or in a virtual atlas (Complete Anatomy). Use an anatomical atlas to locate these structures. If the structure is a foramen, know the major structures passing through it. Skulls will be available in the library or Structure Lab.

- **Internal Basicranium:**
    - **Anterior Cranial Fossa**: cribriform plate
    - **Middle Cranial Fossa:** optic canal, superior orbital fissures, foramen rotundum, ovale, spinosum
    - **Posterior Cranial Fossa**: internal acoustic (auditory) meatus, jugular foramen, foramen magnum, hypoglossal canal
- **External Basicranium/Inferior Aspect of Skull:** styloid process, mastoid process, stylomastoid foramen, jugular foramen
- **Face:** supraorbital, infraorbital, mental, and mandibular foramina